

Immunopathogenesis of vaccine-enhanced RSV disease

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Abstract

Inducing a strong immune response is an essential aim of vaccination. Although immune responses to virus infections are usually protective, they can also be harmful. The best-documented examples of an immune response increasing disease severity are with dengue, measles and respiratory syncytial virus infections.

In the 1960s, administration of formalin-inactivated, tissue culture grown RSV (FI-RSV) was found to induce strong ELISA binding but poor virus-neutralising antibody. Infants given this 'lot 100' vaccine appeared to exhibit an increased rate of RSV infection during subsequent natural RSV outbreaks. Although it has not been possible to exactly delineate the cause of disease enhancement in man, animal models strongly suggest that it was due to strong (and perhaps unbalanced) T cell priming rather than infection-enhancing or sensitising antibody.

In animal models, enhanced disease can result from over-exuberant T cell priming which recruits an abundant inflammatory infiltrate in the lung (the nature of which depends on the patterns of cytokines and chemokines produced). Formalin-treated RSV vaccination has been linked specifically to the induction of Th2 cells, which make IL-4 and IL-5 and induce a strong pulmonary eosinophilic response. The vaccine dosing regime and the interval between vaccination and challenge can be critical to the induction of protection or pathology. Defining the correlates of protection and disease enhancement in man is critical to the rational development of effective and protective vaccines against RSV. © 2001 Published by Elsevier Science Ltd.

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1. Human RSV formalin vaccine trials

Following the success of killed polio vaccine and other inactivated viral vaccines (given singly or in combination), there was hope that other viruses could be chemically inactivated and retain their immunogenic properties. Trials with formalin-inactivated RSV vaccine were therefore conducted in 1966–1967 in infants and children aged 2 months to 9 years. During subsequent RSV exposure, the rate of the virus infection in infants who received the vaccine was no less (and was perhaps even greater) than that in control group immunised with parainfluenza vaccine. Most remarkably, 80% of RSV vaccinees needed hospitalisation, whereas only 5% of such infections among control parainfluenza vaccinees required admission to the hospital [1].

Illnesses among vaccinated children included pneumonia, bronchiolitis, rhinitis or bronchitis. Sadly, two infants died. The major findings at post-mortem examination were extensive bronchopneumonia with emphysema and pneumothorax. Microscopic findings indicated an intense inflammatory

cellular infiltrate in the lungs comprising mononuclear cells, eosinophils and polymorphonuclear cells. RSV could be grown from the lung (at 10^4 TCID₅₀ per gram of tissue). Additionally, pure culture of *Escherichia coli* was recovered from the trachea, lung, blood and spleen of one infant and *Klebsiella* from the lung, trachea and nose culture from the other.

The intense inflammatory infiltrate in the lungs of vaccinated children suggested an immunopathological cause of enhanced disease. Limited data is available regarding the immunological status of vaccinees before, during and after the immunisation. At the time of the study, it was commonly believed that antibodies were responsible for protection and blood samples were collected and stored to study humoral responses. Although there was a significant rise in specific ELISA antibody titres, the levels of neutralising antibodies were poor [1]. Subsequent investigations showed that serum from vaccinated children had poor fusion inhibiting antibody, raising the possibility that fusion protein was altered during formalin treatment [2].

Significantly, the highest incidence of serious disease occurred in children younger than 6 months (when maternally derived serum antibody is still present). It has been suggested that the altered reactivity in the RSV vaccinees was a

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type of delayed hypersensitivity, resulting in enhanced pulmonary infection upon natural exposure.

RSV is not the only example of antiviral immunity enhancing pathology to subsequent infection, which is also seen in dengue and measles. This may be a consequence of the vaccine preparations used. Other trials have reported that children immunised with vaccines, which were formalin or Tween and diethyl ether-inactivated did not acquire protective immune responses [3–5]. Moreover, when the virus became prevalent in the community, they developed either atypical [3] or more severe disease [1].

The vaccine dosing regime and the interval between vaccination and challenge may be critical to the induction of protection or pathology. Indeed, in the case of measles, the enhanced cutaneous disease seen after formalin-inactivated vaccination of humans was delayed up to 7 years, whereas with RSV the enhanced lung disease was seen within 2 years of vaccination. As immunological memory wanes, a window of protection may be followed by a window of disease enhancement [6], followed by a final period during which immune memory is still measurable but neither protection nor disease enhancement is seen.

Research in to RSV vaccines has been coloured by the disastrous consequences of trials of formalin-inactivated RSV (FI-RSV) in the late 1960s. Understanding the mechanism of vaccine-enhanced illness in RSV is an essential step to designing new vaccines for this important infection, and animal models (correctly interpreted and not overused) are helpful in understanding disease augmentation.

2. Animal models for studying immunopathogenesis of vaccine-enhanced RSV disease

Formalin-inactivated preparations of RSV have been used to immunise susceptible animals and characterise immune responses after RSV challenge. In initial studies, Cotton rats were vaccinated with FI-RSV and challenged with RSV intranasally. The lungs of immunised animals were infiltrated with neutrophils, macrophages and lymphocytes and the serum contained only low levels of neutralising antibodies [7]. The authors of these studies suggested that epitopes involved in the production of neutralising antibody could have been modified by formalin treatment and that non-protective epitopes induced high titres of antibodies that form immune complexes. However, similar lung pathology can sometimes be seen when cotton rats are immunised with either purified F protein or chimerical FG protein, and high-dose purified F protein induces elevated levels of anti-RSV antibodies but with low neutralising activity [6,8].

The central role of T cells in augmented lung pathology has been highlighted in the mouse model of RSV. Connors et al. showed that CD4⁺ T cells are crucial to the immunopathogenesis of vaccine-augmented RSV disease, and that RSV specific antibodies (in the absence of CD4⁺ and CD8⁺ T cells) are not sufficient to cause disease

enhancement [9]. Further studies reveal a marked increase in the expression of Th2-type cytokines (IL-5, IL-13, IL-10) and reduction in the expression of IL-12 in FI-RSV immunised mice, indicating a Th2 bias in enhanced inflammation [10]. The same authors found a positive correlation between the signal for IL-5 mRNA and the eosinophil infiltration. Moreover, skewing to a Th1 pattern of cytokine production by priming with live RSV prevented subsequent enhanced disease [11].

We have used a mouse model to further dissect the contribution of different T cell subsets and RSV antigens to the pathology of RSV infection. We have used recombinant vaccinia viruses (mostly kind gifts of Profs. G. Wertz, J. Melero and co-workers) expressing individual RSV proteins to vaccinate mice [12]. We have observed dramatically different outcomes, depending upon the RSV protein used to prime [13]. For example, in BALB/c mice the surface G protein primes for an eosinophilic inflammatory response, reminiscent of the responses seen in the FI-RSV model [14]. This response is mediated by Th2 type CD4⁺ T cells [15]. Using deletion mutations of regions of the G protein, we have identified a single epitope recognised by the Th2 cells [16]. Deletion mutants of this region of G no longer prime an eosinophilic response to infection, but do prime the immune system to clear the virus more effectively. Th2 priming with G protein does not occur in all strains of mice, but arises as a result of a balance between the Th2 T cells and IFN- γ producing CD8⁺ cells. Treatment of otherwise eosinophilia-resistant mice with anti-CD8⁺ antibody (or using mice genetically deficient in functional CD8⁺ cells) allows the development of eosinophilic inflammation in previously resistant strains [17].

Could stimulation of a strong IFN- γ secreting CD8⁺ population of cells be a means of preventing immunopathology induced by RSV vaccination and enhancing viral clearance? Mouse models have again offered insight into the role of CD8⁺ cells in immunopathology. The M2 protein of RSV contains the immunodominant class I epitope for BALB/c mice. Incorporation of the M2 gene into vaccinia, or mucosal vaccination with the peptide epitope in conjunction with the adjuvant LTK63, has allowed us to prime strong CTL responses [18]. Priming, or transfer of CTLs to naïve mice, does indeed result in accelerated viral clearance [19]. However, immunopathology is also greatly enhanced: animals with an unchecked and highly active CTL population experience augmented pathology with severe lung haemorrhage, a neutrophilic infiltrate, and dramatic weight loss, reminiscent of 'shock lung'.

In summary, animal models suggest that the immunopathology seen in FI-RSV vaccines was a result of a Th2 driven eosinophilic inflammation similar to that induced by priming with the major surface glycoprotein G in the absence of F. The rational design of an RSV vaccine will require an understanding of how different immune responses are primed and how different antigens and arms of the immune system are involved in both protective and

immunopathological consequences of vaccination. The challenge lies in inducing a good protective T cell response, whilst avoiding the pathological consequences of unbalanced Th2 or CTL responses. In the light of animal studies demonstrating the role of immune responses in the pathology of RSV infection, human studies of immune correlates with disease will be critical to the understanding of the pathology of RSV in man.

3. Immune correlates of pathology in RSV infection

RSV bronchiolitis is a disease of early life. However, although all of us are exposed to RSV during the first few years of life, only a few infants develop severe respiratory infections. As in the vaccination models, inappropriate immune responses have been implicated in driving pulmonary inflammation during naturally acquired infection.

4. The role of cell mediated immunity in RSV pathology

There is increasing evidence that a heightened cell mediated immune response occurs in infants with severe pulmonary disease, and may contribute to airway injury [20]. Children with impaired CMI show a prolonged shedding of RSV [21] and develop RSV pneumonia at an age when this is rather uncommon (over 3 years old). Chiba et al. found that cellular cytotoxic activity was age-dependent with the response found in 65% of those aged 6–24 months but in only 35–38% of those aged 5 months or less; this may, in part, account for the severity of the disease in the younger age group [22]. The reason for the relatively poor CMI response in the young may be immunological immaturity but it is more likely that the greater response seen in older children is a reflection of previous RSV infection; the enhanced response may simply represent booster effects secondary to stimulation of memory cells.

Using different methodology to that of Chiba et al., others found a significant CMI response in 78% of infants under 6 months compared with 46% over 6 months [20]. The greater CMI response in the age group most severely affected by RSV would indicate that CMI is involved in the pathogenesis of RSV as well as in severe pulmonary disease in infants with natural RSV infection who had previously been immunised with RSV vaccine [23]. It has also been shown that infants who develop RSV-specific CMI within a few days of infection had a clinical illness characterized by wheeze [24] whereas CMI developed more gradually in those with upper respiratory illness or pneumonia. In addition, the infants with a high degree of RSV-specific CMI activity 3–9 weeks after onset of infection were more prone to wheezing over the next 6 months. It seems that although a heightened T cell response is necessary to clear the virus, airway injury occurs as the virus is eradicated. As seen in the animal

models, CMI contributes to host defense against RSV but at the same time causes much of the pathological process.

5. The role of humoral immunity in RSV pathology

The humoral immune response has an important role in protection against respiratory viruses, which is confirmed by the beneficial effect on incidence and severity of RSV bronchiolitis in children given prophylactic RSV immune globulin or humanised monoclonal antibody [25]. In a similar way to the CMI response, age has an important effect on humoral immunity. Infants under 6–8 months mount a poor antibody response to RSV infection compared to older infants, and there is often a failure to develop a protective secretory IgA or neutralizing IgG response [26].

It has been argued that the presence of passively acquired maternal antibody may have an immunosuppressive effect on the development of the infant's own immune response. It seems that it is the response to the G glycoprotein of RSV that is particularly affected by maternal antibodies [27]. There is contrasting evidence as to whether these maternal antibodies are harmful or helpful. It has been hypothesized that due to lack of specific secretory IgA during the initial challenge with RSV, maternal IgG could diffuse to the lumen of the infant airways and form immune complexes [28]. Phagocytosis of these RSV:IgG immune complexes has been shown to stimulate release of inflammatory mediators by neutrophils, which could contribute to the pathological process. Conversely, there is good evidence that maternal antibodies have a protective role in the first month of life and passive antibody therapy is effective in preventing disease in high-risk children [29]. RSV infection is commonest under the age of 6 months when these antibody titres are at their highest. As well as this, infants with the highest titres of transplacentally-acquired RSV-specific IgG tend to develop less severe RSV pneumonia than those with lower titres and have significantly fewer infections [30]. Although high levels of passive antibody are protective, maternal antibody is detectable in infants up to 5 months and yet fails to protect children with the highest incidence of disease. However, the underlying cause of this association remains undetermined.

Abnormal T cell regulatory mechanisms may be related to a hyperactive IgE response. Welliver et al. have shown a reduced number of CD8 positive cells (suppressor/cytotoxic T lymphocytes) during convalescence in those who wheezed with RSV compared to those with upper respiratory illness alone [31]. There was also an inverse correlation between RSV-IgE in nasopharyngeal secretions and CD8⁺ cells in peripheral blood and it is suggested that these cells may include some that are responsible for suppression of IgE production. It is unknown whether this abnormal IgE regulation is virus-induced or constitutionally determined. Host factors, which regulate the response of T lymphocytes to RSV, may be critical in determining the clinical outcome of RSV infection. CD8 T cells and NK cells are an abundant source

of IFN γ , which inhibits the development of CD4 T cells making IL-4 and IL-5, and it is these latter cytokines, which are responsible for IgE production and eosinophilia [32].

Future design of an RSV vaccine will require answers to the following questions. What is the nature of the immune responses associated with detrimental versus beneficial responses in man? Can we associate recognition of particular RSV proteins or epitopes (by T cells or antibodies) with stimulation of particular arms of the immune system, or with different pathologies? Finally, once we have identified the beneficial mechanisms of viral clearance, and those we wish to avoid, how can we stimulate safe and effective immunity in young infants before they encounter natural RSV infection for the first time?

References

- [1] Kim HW, Canchola JG, Brandt CD, Pyles G, Chanock RM, Jensen K, et al. Respiratory syncytial virus disease in infants despite prior administration of antigenic inactivated vaccine. *Am J Epidemiol* 1969;89:422–34.
- [2] Murphy BR, Walsh EE. Formalin-inactivated respiratory syncytial virus vaccine induces antibodies to the fusion glycoprotein that are deficient in fusion-inhibiting activity. *J Clin Microbiol* 1988;26:1595–7.
- [3] Fulginiti VA, Eller JJ, Downie AW, Kempe CH. Altered reactivity to measles virus: atypical measles in children previously immunised with inactivated measles virus vaccines. *JAMA* 1967;202:1075–80.
- [4] Kapikian AZ, Mitchell RH, Chanock RM, Shvedoff RA, Stewart CE. An epidemiologic study of altered clinical reactivity to respiratory syncytial (RS) virus infection in children previously vaccinated with an inactivated RS virus vaccine. *Am J Epidemiol* 1969;89:405–21.
- [5] Norrby E, Gollmar Y. Identification of measles virus-specific haemolysis-inhibiting antibodies separate from haemagglutination-inhibiting antibodies. *Infect Immun* 1975;11:231–9.
- [6] Murphy BR, Sotnikov AV, Lawrence LA, Banks SM, Prince GA. Enhanced pulmonary histopathology is observed in cotton rats immunised with formalin-inactivated respiratory syncytial virus (RSV) or purified F glycoprotein and challenged with RSV 3–6 months after immunisation. *Vaccine* 1990;8:497–502.
- [7] Prince GA, Jensen AB, Hemming VG, Murphy BR, Walsh EE, Horswood RL, et al. Enhancement of respiratory syncytial virus pulmonary pathology in cotton rats by prior intramuscular inoculation of formalin-inactivated virus. *J Virol* 1986;57:721–8.
- [8] Connors M, Collins PL, Firestone CY, Sotnikov AV, Waitze A, Davis AR, et al. Cotton rats previously immunised with a chimerical RSV FG glycoprotein develop enhanced pulmonary pathology when infected with RSV, a phenomenon not encountered following immunisation with vaccinia–RSV recombinants or RSV. *Vaccine* 1992;10:475–84.
- [9] Connors M, Kulkarni AB, Firestone CY, Holmes KL, Morse HC, Sotnikov AV, Murphy BR. Pulmonary histopathology induced by respiratory syncytial virus (RSV) challenge of formalin-inactivated RSV-immunised BALB/c mice is abrogated by depletion of CD4⁺ T cells. *J Virol* 1992;66:7444–51.
- [10] Waris ME, Tsou C, Erdman DD, Zaki SR, Anderson LJ. Respiratory syncytial virus infection in BALB/c mice previously immunised with formalin-inactivated virus induces enhanced pulmonary inflammatory response with a predominant Th2-like cytokine pattern. *J Virol* 1996;70:2852–60.
- [11] Waris ME, Tsou C, Erdman DD, Day DB, Anderson LJ. Priming with live respiratory syncytial virus (RSV) prevents the enhanced pulmonary inflammatory response seen after RSV challenge in BALB/c mice immunised with formalin-inactivated RSV. *J Virol* 1997;71(9):6935–9.
- [12] Openshaw P, Murphy EE, Hosken NA, Maino V, Davis K, Murphy K, et al. Heterogeneity of intracellular cytokine synthesis at the single cell level in polarised T helper 1 and T helper 2 populations. *J Exp Med* 1995;182:1357–67.
- [13] Srikiatkachorn A, Braciale TJ. Virus-specific memory and effector T lymphocytes exhibit different cytokine responses to antigens during experimental murine respiratory syncytial virus infection. *J Virol* 1997;71(1):678–85.
- [14] Openshaw PJM, Clarke SL, Record FM. Pulmonary eosinophilic response to respiratory syncytial virus infection in mice sensitised to the major surface glycoprotein G. *Int Immunol* 1992;4:493–500.
- [15] Alwan WH, Kozłowska WJ, Openshaw PJM. Distinct types of lung disease caused by functional subsets of antiviral T cells. *J Exp Med* 1994;179:81–9.
- [16] Sparer TE, Matthews S, Hussell T, Rae AJ, García-Barreno B, Melero JA, et al. Eliminating a region of respiratory syncytial virus attachment protein allows induction of protective immunity without vaccine-enhanced lung eosinophilia. *J Exp Med* 1998;187(11):1921–6.
- [17] Hussell T, Baldwin CJ, O'Garra A, Openshaw PJM. CD8⁺ T cells control Th2-driven pathology during pulmonary respiratory syncytial virus infection. *Eur J Immunol* 1997;27(12):3341–9.
- [18] Simmons CP, Hussell T, Sparer T, Walzl G, Openshaw P, Dougan G. Mucosal delivery of a respiratory syncytial virus CTL peptide with enterotoxin-based adjuvants elicits protective, immunopathogenic, and immunoregulatory antiviral CD8⁺ T cell responses. *J Immunol* 2001;166(2):1106–13.
- [19] Cannon MJ, Openshaw PJM, Askonas BA. Cytotoxic T cells clear virus but augment lung pathology in mice infected with respiratory syncytial virus. *J Exp Med* 1988;168:1163–8.
- [20] Scott R, Kaul A, Scott M, Chiba Y, Ogra PL. Development of in vitro correlates of cell-mediated immunity to respiratory syncytial virus infection in humans. *J Infect Dis* 1978;137:810–7.
- [21] Fishaut M, Tubergen D, McIntosh K. Cellular response to respiratory viruses with particular reference to children with disorders of cell mediated immunity. *J Pediatr* 1980;96:179–86.
- [22] Chiba Y, Higashidate Y, Suga K, Honjo K, Tsutsumi H, Ogra PL. Development of cell-mediated cytotoxic immunity to respiratory syncytial virus in human infants following naturally acquired infection. *J Med Virol* 1989;28:133–9.
- [23] Kim HW, Leikin SL, Arrobio J, Brandt CD, Chanock RM, Parrott RH. Cell mediated immunity to respiratory syncytial virus induced by inactivated vaccine or by infection. *Pediatr Res* 1976;10:75–8.
- [24] Welliver RC, Kaul A, Ogra PL. Cell-mediated immune response to respiratory syncytial virus infection: relationship to the development of reactive airway disease. *J Pediatr* 1979;94:370–5.
- [25] Groothuis JR, Simoes EA, Levin MJ, Hall CB, Long CE, Rodriguez WJ, et al. Prophylactic administration of respiratory syncytial virus immune globulin to high-risk infants and young children. The Respiratory Syncytial Virus Immune Globulin Study Group. *New Engl J Med* 1993;329 (21):1524–30.
- [26] Murphy BR, Graham BS, Prince GA, Walsh EE, Chanock RM, Karzon DT, et al. Serum and nasal-wash immunoglobulin G and A antibody response of infants and children to respiratory syncytial virus F and G glycoproteins following primary infection. *J Clin Microbiol* 1986;23:1009–14.
- [27] Murphy BR, Alling DW, Snyder MH, Walsh EE, Prince GA, Chanock RM, et al. Effect of age and pre-existing antibody status on serum antibody responses of infants and children to the F and G glycoproteins during respiratory syncytial virus infection. *J Clin Microbiol* 1986;24:894–8.
- [28] Nadal D, Ogra PL. Development of local immunity: role in mechanisms of protection against or pathogenesis of respiratory syncytial viral infections. *Lung* 1990;168(Suppl):379–87.

- [29] Groothuis JR. Treatment and prevention of severe respiratory syncytial virus infection in young children. *Curr Opin Infect Dis* 1995;8:206–8.
- [30] Glezen WP, Paredes A, Allison JE, Taber LH, Frank AL. Risk of respiratory syncytial virus infection for infants from low-income families in relationship to age, sex, ethnic group, and maternal antibody level. *J Pediatr* 1981;98:708–15.
- [31] Welliver RC, Kaul TN, Sun M, Ogra PL. Defective regulation of immune responses in respiratory syncytial virus infection. *J Immunol* 1984;133:1925–30.
- [32] Openshaw PJM. Immunity and immunopathology to respiratory syncytial virus: the mouse model. *Am J Respir Crit Care Med* 1995;152:S59–62.